

YAN DAI

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EDUCATION

Massachusetts Institute of Technology Operations Research Center 2024 — Now

PhD student in Operations Research co-advised by Patrick Jaillet and Negin Golrezaei. GPA: 5.0 / 5.

Tsinghua University Andrew C. Yao Class 2020 — 2024

Bachelor of Engineering in Computer Science and Technology. GPA: 3.99/4.00.

RESEARCH EXPERIENCES

Microsoft Research Economics and Computer Science Group (Cambridge, MA) May 2026 — Now

Research intern of Alex Silvikins and Brendan Lucier (also working closely with Nicole Immorlica).

Massachusetts Institute of Technology OR Center (Cambridge, MA) Sep. 2024 — Now

Research assistant of Patrick Jaillet and Negin Golrezaei.

Tsinghua University Andrew C. Yao Class (Beijing, China) Feb. — Jun. 2024

Bachelor's thesis advised by Longbo Huang.

University of Washington School of CSE (Seattle, WA) Jun. — Aug. 2023

Visiting student hosted by Simon S. Du.

Massachusetts Institute of Technology Lab for Info. & Decision (Cambridge, MA) Feb. — May 2023

Visiting student hosted by Suvrit Sra.

University of Southern California CS Department (Remote) Feb. — Dec. 2022

Remote visitor of Haipeng Luo.

Tsinghua University Andrew C. Yao Class (Beijing, China) Jun. — Dec. 2021

Research assistant of Longbo Huang.

PUBLICATIONS

(* stands for equal contribution. Listed reverse chronologically within each category.)

Economics and Computer Science

[15] [Yan Dai](#), Maryam Farboodi, Negin Golrezaei, and Sepehr Shahshahani. “Market Design for AI: Beyond the Copyright Binary.”

[14] **[NeurIPS’25]** [Yan Dai](#), Negin Golrezaei, and Patrick Jaillet. “Incentive-Aware Dynamic Resource Allocation under Long-Term Cost Constraints.” [Preprint]

1st Place in ACM Student Research Competition (SRC) @ SIGMETRICS’25.

[13] **[COLT’25]** [Yan Dai](#), Moïse Blanchard, and Patrick Jaillet. “Non-Monetary Mechanism Design without Priors: Achieving Efficiency via Adaptive Costly Audits.” [Preprint] [Proceeding (Extended Abstract)]

Bandits and Online Learning

[12] [Yan Dai](#), Negin Golrezaei, and Patrick Jaillet. “Policy Regret for Embedding Model Routing: Contextual Bandits with Low-Rank Experts.”

[11] **[ACM POMACS & SIGMETRICS’25]** [Yan Dai](#) and Longbo Huang. “Adversarial Network Optimization under Bandit Feedback: Maximizing Utility in Non-Stationary Multi-Hop Networks.” In *Proceedings of the ACM on Measurement and Analysis of Computing Systems* (ACM POMACS), 8(3):31, 2024. [ACM Journal Page]

Best Paper Award of ACM SIGMETRICS’25.

- [10] **[ICLR'25]** Yu Chen*, Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs.” *Spotlight (5%)*. [Preprint] [OpenReview]
- [9] **[ICML'23]** Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “Banker Online Mirror Descent: A Universal Approach for Delayed Online Bandit Learning.” [Preprint] [Proceeding]
- [8] **[ICLR'23]** [Yan Dai](#), Ruosong Wang, and Simon S. Du. “Variance-Aware Sparse Linear Bandits.” [Preprint] [OpenReview]
- [7] **[ICML'22]** Jiatai Huang*, [Yan Dai*](#), and Longbo Huang. “Adaptive Best-of-Both-Worlds Algorithm for Heavy-Tailed Multi-Armed Bandits.” [Preprint] [Proceeding]

Reinforcement Learning Theory

- [6] Aarush Kulkarni, Khang Nguyen, Ricardo Parada, Kenny Guo, William Chang, and [Yan Dai](#). “Learning Adversarial Continuous MDPs with Bandit Feedback and Unknown Transitions.”
- [5] **[COLT'24]** [Yan Dai](#), Qiwen Cui, and Simon S. Du. “Refined Sample Complexity for Markov Games with Independent Linear Function Approximation.” [Preprint] [Proceeding (Extended Abstract)]
- [4] **[ICML'23]** [Yan Dai](#), Haipeng Luo, Chen-Yu Wei, and Julian Zimmert. “Refined Regret for Adversarial MDPs with Linear Function Approximation.” [Preprint] [Proceeding]
- [3] **[NeurIPS'22]** [Yan Dai](#), Haipeng Luo, and Liyu Chen. “Follow-the-Perturbed-Leader for Adversarial Markov Decision Processes with Bandit Feedback.” [Preprint] [OpenReview] [Proceeding]

Deep Learning Theory

- [2] **[ICML'24]** Kwangjun Ahn, Zhiyu Zhang, Yunbum Kook, and [Yan Dai](#). “Understanding Adam Optimizer via Online Learning of Updates: Adam is FTRL in Disguise.” [Preprint] [Proceeding]
- [1] **[NeurIPS'23]** [Yan Dai*](#), Kwangjun Ahn*, and Suvrit Sra. “Curious Role of Normalization in Sharpness-Aware Minimization.” [Preprint] [OpenReview] [Proceeding]

RESEARCH AWARDS AND SCHOLARSHIPS

Research Awards

- Best Paper Award (*1 paper per year*) of ACM SIGMETRICS'25. Jun. 2025
- 1st Place in ACM Student Research Competition (SRC) @ SIGMETRICS'25. Jun. 2025

Fellowship and Scholarship

- Runner-Up of Two Sigma PhD Fellowship (*2 winners and 2 runners-up*). Feb. 2026
- Outstanding Graduate named by Beijing (5%), by Tsinghua (2%), and by Yao Class (10%). Jun. 2024
- Presidential Scholarship (*top scholarship for Tsinghua undergrads; 10/16,000+*), Tsinghua. Nov. 2023
- Andrew C. Yao Award – Gold Medal (*top scholarship for seniors; 1 student*), Yao Class. Sep. 2023

Competitive Programming

- Gold Medal (*1st place with perfect score*), Asia-Pacific Informatics Olympiad (APIO). May 2019
- Gold Medal (*5th place*), Chinese National Olympiad in Informatics (NOI). Jul. 2018
- Gold Medal, Asia-Pacific Informatics Olympiad (APIO). May 2018
- Gold Medal, IOI China Team Selection Competition (CTSC). May 2018
- Gold Medal, Chinese Olympiad in Informatics Winter Camp (WC). Feb. 2018

Miscellaneous

- 1st Place, MIT Intramurals League on Water Polo. Apr. 2023
- 2nd Place, National Collegiate Water Polo Championships. Jun. 2021
- 1st Place, Tsinghua Swimming Competition on Men's 100m Butterfly. Nov. 2021
- 3rd Place, Tsinghua Swimming Competition on Men's 200m Medley. Nov. 2021

PROFESSIONAL SERVICES

Journal Reviewing. Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Machine Learning (ML); IEEE/ACM Transactions on Networking (TON), Performance Evaluation (PEVA), IEEE Internet of Things Journal (IoT-J).

Conference Reviewing. COLT 2025; ICML 2026 (Gold reviewer) / 2025 / 2023; NeurIPS 2026 / 2025 / 2024 / 2023 / 2022; ICLR 2025 / 2024; AISTATS 2026 / 2025 / 2023 / 2022; AAAI 2026; ALT 2023.